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(54) Title: CEMENTITIOUS TILE ADHESIVE COMPOSITION COMPRISING BIOCHAR

(57) Abstract: The present invention relates to a cementitious tile adhesive composition comprising biochar in a specific weight ratio to the cementitious binder or, where the cementitious binder comprises slag, in a specific weight ratio to slag. The present invention also relates to the preparation method of a cementitious tile adhesive composition with a reduced carbon footprint, especially a carbon-negative cementitious tile adhesive composition.

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CEMENTITIOUS TILE ADHESIVE COMPOSITION COMPRISING BIOCHAR

Technical Field

The present invention relates to a cementitious tile adhesive composition comprising biochar. The present invention also relates to the preparation method of a cementitious tile adhesive composition with a reduced carbon footprint, especially a carbon-negative cementitious tile adhesive composition.

Background of the invention

In recent years, the construction industry has dedicated substantial resources to investigating methods for decarbonizing its products and processes. To diminish the carbon footprint of their formulations, mortar developers and concrete manufacturers seek raw materials with a reduced carbon footprint or, ideally, materials with carbon-sinking potential. What is meant by carbon-sinking potential is that these raw materials absorb more CO_2 in their lifetime than they release to the atmosphere. Biochar is an example of material with a carbon-sinking potential.

One specific example of carbon footprint reduction in the construction industry is the use of tile adhesives with a reduced global warming potential (GWP). Currently, such adhesives are being developed by, at least partially, replacing Ordinary Portland Cement with alternative binders having a lower carbon footprint, such as fly ash, slag, silica fume, or metakaolin.

Cementitious compositions comprising biochar are already known (US2023002276, HeroSand, LLC; DE102022002721, Körning H.O., Preuss A.; EP4357312, Saint-Gobain Weber France). Such cementitious compositions have a reduced carbon footprint. However, a simple substitution of either a filler, aggregate or a binder with biochar is likely to lead to a reduced performance such as deteriorated workability, strength, and/or adhesion. When formulating a cementitious tile adhesive composition, only a specific selection and combination of Portland cement, supplementary cementitious materials (SCMs) and biochar can result in a cementitious tile adhesive that meets the requirements for its intended use. This is particularly relevant when formulating a tile adhesive composition for tiles with a low

water absorption (≤ 0.5 %, measured according to EN ISO 10545-3:2018), for example porcelain tiles.

Therefore, there is still a need for improved cementitious tile adhesive compositions with a reduced carbon footprint, most preferably with a negative carbon footprint.

Summary of the invention

It is the objective of the present invention to provide a cementitious tile adhesive composition with a decreased carbon footprint. More preferably, the present invention provides a carbon-negative cementitious tile adhesive composition. Also preferably, the cementitious tile adhesive composition does meet relevant performance requirements, for example for tile adhesives.

It has surprisingly been found that the objective of the present invention is solved by the subject matter of claim 1. It is thus at the core of the present invention to use biochar, in some embodiments in combination with slag, to reduce the carbon footprint of a cementitious tile adhesive composition.

Further aspects of the present invention are the subject of independent claims. Preferred embodiments are the subject matter of dependent claims.

Ways to carry out the invention

In a first aspect the present invention relates to a cementitious tile adhesive, comprising:

a) at least one cementitious binder,

b) at least one aggregate,

c) biochar,

wherein, if the cementitious binder does not comprise slag, the weight ratio of biochar to the cementitious binder is between 0.001 and 0.15 to 1, preferably between 0.001 and 0.024 to 1, more preferably between 0.001 and 0.020 to 1, most preferably between 0.001 and 0.014 to 1, or

wherein, if the cementitious binder comprises slag, the weight ratio of biochar to slag is between 0.16 and 0.5 to 1, preferably between 0.16 and 0.3 to 1, more preferably between 0.16 and 0.25 to 1, especially between 0.16 and 0.2 to 1.

In this context, "tiles" are in particular sheets of wear-resistant materials such as ceramics, stones, metals and/ or glass, preferably ceramic tiles according to 14411:2016. Typically, tiles are used to cover roofs, floors, walls and/ or showers.

A cementitious tile adhesive of the present invention in particular fulfills the requirements of standard EN 12004-1:2017-05.

In particular, the cementitious tile adhesive composition is present as a dry mixture. This means that the cementitious tile adhesive composition is substantially free of water or comprises water in an amount of less than 1 wt. %, in particular less than 0.5 wt. % or less than 0.1 wt. %, relative to the total dry weight of the cementitious tile adhesive composition.

According to preferred embodiments, the cementitious tile adhesive composition is a one component mixture. This means that all individual materials and/ or substances are mixed. One-component compositions are particularly easy to handle and eliminate the risk of user confusion or mis-dosing of the various components.

A cementitious binder, in the context of the present invention, is a binder, which in the presence of water reacts in a hydration reaction to form solid hydrates or hydrate phases. This can be, for example, a hydraulic binder (e.g. cement or hydraulic lime), a latent hydraulic binder (e.g. slag), a pozzolanic binder (e.g. fly ash) or a nonhydraulic binder (gypsum plaster or white lime).

According to embodiments, the cementitious binder is a composite cement of the type CEM II, CEM III, CEM IV or CEM V according to EN 197-1:2011, or is a limestone calcined clay cement.

Within the present context, the Portland cement, also called Ordinary Portland Cement (OPC), is a CEM I under standard DIN 197-1:2011. However, other OPC classified, for example, under the relevant ASTM, JIS or Chinese standards is also suitable. According to further embodiments, the OPC is a white cement. White cements can be preferred within the present context as they have a lower water demand.

Limestone calcined clay cement (LC3) is known to the person skilled in the art and described, for example, in the review paper “Research evolution of limestone calcined clay cement (LC3), a promising low-carbon binder – A comprehensive overview” by J. Mañosa, A. Calderón, R. Salgado-Pizarro, A. Maldonado-Alameda, J.

M. Chimenos, (Vol. 10(3), 2024, Heliyon). Examples of LC3 binder-type compositions are [Portland Cement wt. %: Calcined Clay wt. %: Limestone wt. %] = [70:20:10], [55:30:15], [40:40:20].

As known to the person skilled in the art, CEM II comprises granulated blast furnace slag.

According to embodiments, the cementitious binder comprises Portland cement and slag, preferably steelmaking slag or ground granulated blast furnace slag

According to preferable embodiments, the granulated blast furnace slag is a ground granulated blast furnace slag (GGBS).

Slag, within the context of the present invention, especially is iron making slag and/or steelmaking slag. One type of iron making slag is GGBS (also called GGBFS). A ground granulated blast furnace slag (GGBS) within the present context is obtained by quenching molten iron slag from a blast furnace in water or steam, to produce a glassy, granular product that is then dried and ground into a fine powder.

According to further preferable embodiments, GGBS has a Blaine fineness of between 2000 – 12000 cm^2/g, preferably between 4000 – 6000 cm^2/g or 6000 – 8000 cm^2/g. Blaine fineness can be measured according to standard EN 196-6:2018.

According to embodiments, the cementitious binder is a mixture of a Portland cement and steelmaking slag, preferably basic oxygen furnace slag, ladle slag, or electric arc furnace slag.

Steel making slag within the present context is a by-product from the steelmaking process. Steel making slag is obtained for example in the Thomas process, the Linz-Donawitz process, the Siemens-Martin process or the electric arc furnace when iron is converted to steel. A very preferred type of steel making slag within the present context is basic oxygen furnace slag (BOF), also called basic oxygen slag (BOS).

According to embodiments, the slag is a blast furnace slag, most preferably granulated blast furnace slag or ground granulated blast furnace slag.

According to embodiments, the weight ratio of Portland cement to slag is between 0.2 and 2 to 1, preferably between 0.4 and 1.2 to 1, more preferably between 0.4 and 1 to 1, still more preferably between 0.8 and 0.99 to 1, especially between 0.8 and 0.9 to 1.

Biochar, in the present context, is a carbon-rich product that remains after biomass is thermodynamically converted in the absence or limited presence of oxygen (a process known as pyrolysis). Through pyrolysis, the carbon stored in the biomass becomes locked in the biochar. The biomass used to produce biochar of the present invention is preferably sourced from biomass waste, for example fruits pits, coconuts shells, common wood chips, waste wood, washed coffee husks, coffee pomace, lemon pomace, sliced hemp straw, paper waste, chicken droppings, hemp clay mixture, hemp waste or mixed waste.

According to embodiments, the biochar is obtained by pyrolysis, preferably by slow pyrolysis, of woody biomass, herbaceous biomass, agricultural biomass, waste biomass, or manure biomass.

Slow pyrolysis, in the present context, relates to a pyrolysis process which operates at lower heating rates and longer residence times, typically between hours to days, than fast pyrolysis. Therefore, it favors the production of solid biochar. This process involves the gradual heating of the feedstock to temperatures ranging from 300 °C to 700 °C, ensuring a more controlled and complete conversion of the material. Slow pyrolysis is recognized for its ability to enhance carbon sequestration.

Slow pyrolysis can be performed in specialized reactors designed for thermochemical conversion, such as fixed-bed, rotary kiln, or auger reactors, which are suitable for processing various types of biomass or organic waste materials.

According to embodiments, the biochar is obtained by pyrolysis at the temperature of at least 400 °C, preferably at least 500 °C, more preferably at least 600 °C, most preferably at least 700 °C.

Pyrolysis temperature is the key factor determining the surface area and porosity of biochar. Higher pyrolysis temperatures result in biochar with increased porosity and larger surface area, enhancing its effectiveness for various applications, such as carbon sequestration.

According to preferable embodiments, the biochar has a density which is lower than a density of a standard cementitious tile adhesive. In particular, the biochar has a density which is lower than the density of the cementitious tile adhesive but without biochar. Therefore, cementitious tile adhesive compositions comprising biochar result in a higher surface coverage than the same compositions but without biochar. The

density of biochar can be measured by helium pycnometry as described in the standard ISO 12154:2014.

A preferred biochar of the present invention has a density of between 1.0 – 2.0 g/cm^3, preferably 1.3 – 1.8 g/cm^3, measured by helium pycnometry.

Biochar, within the context of the present invention, can be seen as a lightweight filler.

According to embodiments, the biochar has an average particle size D50 of more than 0.1 mm, as determined by sieve analysis according to EN 12192-1:2002 or EN 933-1:2012.

According to embodiments, the biochar has an average particle size D50 between 0.0001 mm and 0.5 mm, preferably between 0.1001 mm and 0.5 mm, more preferably between 0.1001 mm and 0.4 mm, most preferably between 0.2 mm and 0.4 mm, as determined by sieve analysis according to EN 12192-1:2002 or EN 933-1:2012.

The particle size D50 is the particle size where 50 w% of an ensemble of particles are smaller.

Biochar with an average particle size D50 between 0.1001 mm and 0.5 mm, most preferably between 0.1001 mm and 0.4 mm, might be advantageous over biochar with a smaller particle size, as it reduces dust generation, making handling and mixing of the cementitious tile adhesive composition safer and cleaner. Additionally, smaller particles of biochar require more processing steps, leading to increased energy consumption and potentially higher costs, which can contribute to a negative environmental impact compared to larger particle sizes. Therefore, opting for larger particle sizes can offer both practical and environmental benefits.

According to embodiments, the surface of biochar is treated, preferably functionalized.

Surface treatment, in the present context, refers to various processes and techniques applied to modify the surface properties of biochar to enhance its performance for specific applications. The primary objectives of surface treatment are to increase the adsorption capacity and improve the stability of biochar, as well as introduce functional groups that can interact with other substances. Examples of common

surface treatment methods include chemical activation, physical activation, biological treatment, surface coating and functionalization.

According to embodiments, surface treatment of biochar involves surface coating with mineral oil. Such treatment leads to reduced dust generation.

Functionalization, in the present context, involves chemically or physically modifying the surface of biochar to introduce specific functional groups (for example carboxyl, hydroxyl, amine, thiol) or generate molecular dipoles that enhance its reactivity and adsorption capacity for target molecules. This process aims to tailor the biochar's properties to improve its performance, for example increase the mechanical strength of the composite, reduce dust formation, further improve electrical conductivity, increase fire resistance and/ or improve thermal conductivity. Functionalization can be achieved through treatments with acids, bases, oxidizing agents, or other chemical reagents. For example, the surface of biochar can be functionalized with silane coupling agents, ionic surfactants and/ or nonionic surfactants.

According to embodiments, the aggregate is selected from at least one of limestone, granite, marble, basalt, olivine, aluminum oxide, sand or a combination thereof, preferably marble and/ or limestone and/ or sand.

The term aggregate as used in the context of the present invention refers to mineral materials that are non-reactive in the hydration reaction of cementitious binders. Aggregates can be any aggregate typically used for cementitious materials such as concrete, mortars, screeds, renders, grouts, coatings, putties or the like. Typical aggregates are for example rock, crushed stone, gravel, slag, limestone, sand, recycled concrete, perlite or vermiculite.

The average particle size of the aggregate is preferably between 0.063 mm and 4 mm, more preferably between 0.063 mm and 2 mm, even more preferably between 0.063 mm and 1 mm, as determined by the sieve analysis according to EN 12192-1:2002 or EN 933-1:2012. However, other aggregates can also be used for specific purposes.

According to embodiments, the cementitious tile adhesive composition further comprises at least one alkaline activator, preferably alkali metal carbonate, alkali metal hydroxide, alkaline-earth metal hydroxide, alkaline-earth metal oxide, alkaline-earth metal formate, alkali metal silicate, alkali metal sulphate, alkali metal chloride or mixture thereof.

According to preferable embodiments, the activator comprises quick lime.

According to embodiments, the cementitious tile adhesive composition further comprises at least one admixture, wherein said admixture is selected from plasticizers, thickeners, retarders, air-entrainers, accelerators, anti-caking agents, antioxidants, de-aerating agents, defoamers, corrosion inhibitors, expansion producing additives, pigments, strength enhancers, waterproofing additives, alkali-aggregate reaction inhibitors, chromate reducers, de-dusting agents, water retention agents, fibers, and/or anti-microbial agents.

Suitable thickeners can be any known to the person skilled in the art. Examples include cellulose ethers, polyacrylamides and alkali swellable polymers.

According to further embodiments, the cementitious tile adhesive composition further comprises a cellulose ether selected from the group consisting of hydroxymethylcellulose, hydroxyethylcellulose, hydroxypropylmethylcellulose, hydroxyethylmethylcellulose and / or carboxymethyl cellulose, preferably hydroxypropyl methylcellulose and / or hydroxyethyl methylcellulose.

According to a preferred embodiment, the amount of cellulose ether is from 0.05 to 3 wt. %, in particular from 0.1 to 1 wt. %, especially from 0.2 to 0.6 wt. %, relative to the total dry weight of the cementitious tile adhesive composition.

According to further embodiments, the cementitious tile adhesive composition further comprises a de-dusting agent. De-dusting agents for cementitious tile adhesives are used to reduce airborne dust during mixing and application, enhancing worker safety and site cleanliness. These agents also help improve the workability and consistency of the adhesive, ensuring a more efficient and effective installation process. Preferably, such de-dusting agent is refined paraffin wax.

According to preferred embodiments, a cementitious tile adhesive composition comprises, relative to the total dry weight of the cementitious tile adhesive composition:

a) 5 to 30 wt. %, preferably 10 to 20 wt. %, most preferably 15 to 20 wt. % of Portland cement;

b) 10 to 30 wt. %, preferably 20 to 30 wt. %, most preferably 20 to 25 wt. % of slag;

b) 40 to 70 wt. %, preferably 45 to 60 wt. %, most preferably 45 to 50 wt. % of at least one aggregate;

c) 0.5 to 10 wt. %, preferably 2 to 8 wt. %, most preferably 3 to 6 wt. % of biochar.

According to further preferred embodiments, a cementitious tile adhesive composition comprises:

a) Portland cement;

b) slag;

b) at least one aggregate;

c) biochar,

wherein the bulk density ratio of Portland cement and slag to aggregates is between 7:3 to 3:7, wherein the bulk density is measured with the apparatus described in DIN EN 459-2:2021, chapter 7.3, “Einlaufgerät nach Böhme”.

According to preferred embodiments, a cementitious tile adhesive composition, comprises, relative to the total dry weight of the cementitious tile adhesive composition:

a) 10 to 45 wt. % of at least one cementitious binder;

b) 50 to 90 wt. % of at least one aggregate;

c) 0.15 to 2.25 wt. % of biochar,

wherein the biochar has an average particle size D50 of more than 0.1 mm, as determined by sieve analysis according to EN 12192-1:2002 or EN 933-1:2012.

According to preferred embodiments, a cementitious tile adhesive composition of the present invention comprises Portland cement and slag, wherein the weight ratio of biochar to slag is between 0.16 and 0.25 to 1, and wherein the weight ratio of Portland cement to slag is between 0.8 and 0.9 to 1.

According to preferred embodiments, a cementitious tile adhesive composition of the present invention has a reduced carbon footprint in comparison to the same cementitious tile adhesive composition, having more aggregates instead of biochar.

Carbon footprint refers to the total amount of greenhouse gases, primarily carbon dioxide, emitted directly or indirectly by a product or process, measured over its

entire lifecycle. Carbon footprint can, for example, be calculated as CO_2 equivalents (CO_2 eq.) per kilogram.

The Global Warming Potential (GWP) was developed to enable the comparison of the global warming effects of various gases. It quantifies how much energy the emissions of 1 ton of a particular gas will absorb over a specified period, compared to the emissions of 1 ton of CO_2. In other words, GWP is a metric used within various carbon footprint measurement methods to compare the impacts of different greenhouse gases.

According to embodiments, the initial tensile adhesion strength of such cementitious tile adhesive composition measured according to DIN EN 12004-1:2017 is at least 0.5 N/mm^2, preferably at least 1 N/mm^2, most preferably at least 1.2 N/mm^2.

In another aspect the present invention relates to a processable composition comprising the cementitious tile adhesive composition according to the present invention and water, wherein the weight ratio of water to cementitious tile adhesive composition is in the range from 0.2 to 0.5, preferably from 0.3 to 0.45.

Processable composition, in the present context, means a cementitious tile adhesive composition that is ready to be applied to a support element, such as a floor structure, to secure a cover element, such as a tile. In particular, the processable composition is in the initial stage of curing of the cementitious binder.

In particular, the processable composition has an air content from 0 to 5%, in particular from 0.1 to 3%. For example, the air content is measured according to DIN EN 1015-7:1998.

Preferably, the processable composition is applied at a thickness ranging from 0.5 to 15 mm, more preferably from 0.5 to 10 mm, in particular from 3 to 6 mm.

All features and embodiments as described above, also apply to this aspect.

In another aspect the present invention relates to the use of the cementitious tile adhesive composition or processable composition of the present invention for tiles with surface dimensions of at least 300 x 300 mm, preferably at least 600 x 600 mm

and/or with a water absorption of ≤ 0.5 % measured according to EN ISO 10545-3:2018, for example porcelain tiles.

The use for tiles, in the present context, means that the tiles are adhered to a surface by a cementitious tile adhesive.

Large format tiles, typically at least 300 × 300 mm in size, are commonly used in both residential and commercial settings. They are particularly popular in modern interior design for flooring and wall application. Their larger size can create a seamless and visually appealing look while reducing grout lines for a cleaner aesthetic. However, large format tiles require a tile adhesive with a tensile adhesion of at least 1 N/mm^2.

All features and embodiments as described above, also apply to this aspect.

In another aspect the present invention relates to a structure, in particular a floor, a wall or a ceiling, comprising a cover element, in particular a tile, whereby the cover element is fixed to the structure with a hardened cementitious tile adhesive composition according to the present invention.

All features and embodiments as described above, also apply to this aspect.

In another aspect the present invention relates to the method for preparing a cementitious tile adhesive composition with a Global Warming Potential of biogenic CO_2 emission of less than 0.07 kg CO_2 eq./ kg, preferably less than 0.065 kg CO_2 eq./ kg, more preferably less than 0.05 kg CO_2 eq./ kg, more preferably less than 0.04 kg CO_2 eq./ kg, especially having a negative Global Warming Potential of biogenic CO_2 emissions, said method comprising a step of adding biochar to the cementitious tile adhesive composition.

The Global Warming Potential (GWP) was developed to enable the comparison of the global warming effects of various gases. It quantifies how much energy the emissions of 1 ton of a particular gas will absorb over a specified period, compared to the emissions of 1 ton of CO_2. In other words, GWP is a metric used within various carbon footprint measurement methods to compare the impacts of different greenhouse gases.

Carbon footprint refers to the total amount of greenhouse gases, primarily carbon dioxide, emitted directly or indirectly by a product or process, measured over its entire lifecycle. Carbon footprint can, for example, be calculated as CO_2 equivalents ($CO_{2\ eq.}$) per kilogram.

In particular, in a method of the present invention, the GWP of a cementitious tile adhesive composition, especially a cementitious tile adhesive composition, is reduced as compared to the same composition but without biochar. In other words, the reduction of GWP of the cementitious tile adhesive composition is achieved by the presence of biochar in the adhesive composition.

In the method of the present invention, biochar is added to a cementitious tile adhesive composition to reduce the GWP below 0.07 kg CO_2 eq./ kg, preferably below 0.065 kg CO_2 eq./ kg, more preferably below 0.05 kg CO_2 eq./ kg, still more preferably below 0.04 kg CO_2 eq./ kg, especially to a negative Global Warming Potential of biogenic CO_2 emissions. If, after adding biochar, the GWP is not reduced below a certain value, more biochar is added. The biochar is added as long as the GWP does not decrease below 0.07 kg CO_2 eq./ kg, preferably below 0.065 kg CO_2 eq./ kg, more preferably below 0.05 kg CO_2 eq./ kg, still more preferably below 0.04 kg CO_2 eq./ kg especially to a negative Global Warming Potential of biogenic CO_2 emissions.

In one of the embodiments, biochar and at least one supplementary cementitious material (SCM) are added to a cementitious tile adhesive composition to reduce the GWP below 0.07 kg CO_2 eq./ kg, preferably below 0.065 kg CO_2 eq./ kg, more preferably below 0.05 kg CO_2 eq./ kg, still more preferably below 0.04 kg CO_2 eq./ kg, especially to a negative Global Warming Potential of biogenic CO_2 emissions. If, after adding biochar and at least one supplementary cementitious material, the GWP is not reduced below a certain value, more biochar and supplementary cementitious material are added. The biochar and at least one supplementary cementitious material are added as long as the GWP does not decrease below 0.07 kg CO_2 eq./ kg, preferably below 0.065 kg CO_2 eq./ kg, more preferably below 0.05 kg CO_2 eq./ kg, still more preferably below 0.04 kg CO_2 eq./ kg, especially to a negative Global Warming Potential of biogenic CO_2 emissions.

Within the present context, supplementary cementitious materials (SCMs) are materials that, when used in combination with Portland cement, contribute to the

properties of the hardened cementitious composition through hydraulic or pozzolanic activity or both. SCMs are often used to replace a portion of the Portland cement in cementitious compositions. They help to reduce the carbon footprint of cementitious composition by reducing the amount of clinker required for cement production. The most commonly used SCMs are fly ash, blast furnace slag, silica fume and clay.

According to preferable embodiments, the cementitious tile adhesive composition has a negative Global Warming Potential of biogenic CO_2 emission.

A negative Global Warming Potential, in the present context, means that through its entire lifecycle the adhesive removes more CO_2 from the atmosphere than it emits or generates, resulting in a net reduction of CO_2 levels.

All features and embodiments as described above, also apply to this aspect.

Experimental part

Example 1

Table 1 shows three cementitious tile adhesive compositions. The compositions have been prepared by intermixing all of the components in a dry state. The adhesive compositions 1-1 to 1-3 are present as a dry powder.

Table 1: Tile adhesives formulations; all values given in [g]. Examples 1-1 (not inventive) and 1-2, 1-3 (inventive).

Example	**1-1**	**1-2**	**1-3**
CEM I, 52.5 R	4.125	4.125	4.125
GGBS	12.375	12.375	12.375
$CaCO_3$	5	5	5
Marble aggregate, < 1 mm	78.2	78.2	78.2
Thickener	0.3	0.3	0.3
Biochar (0.1 – 0.5 mm)	-	0.165	0.33

Tile adhesive compositions 1-1 to 1-3 have been mixed with water (weight ratio of water to total weight of dry adhesive composition was 0.22) in order to obtain processable compositions.

The obtained processable compositions were then used for adhesion tests according to Annex ZA, Table ZA.1.1 of EN 12004-1/2:2017. The results of adhesion tensile strength after (a) dry and (b) wet storage are presented in Table 2.

The carbon footprint of the adhesive compositions disclosed in Table 1 was assessed based on the carbon sequestration potential of 2.05 g of CO_2 per g of biochar. The Global Warming Potential (GWP) values of these adhesives, including biogenic emissions, are presented in Table 2.

Table 2: Results of tested tile adhesive specimens. Examples 1-1 (not inventive) and 1,2, 1-3 (inventive).

Example	**1-1**	**1-2**	**1-3**
Adhesion tensile strength – dry [N/mm^2]	0.51	0.54	0.61
Adhesion tensile strength – wet [N/mm^2]	0.69	0.65	0.82
GWP [kg $CO_{2\ eq.}$/ kg product]	0.071	0.069	0.064

Both tested adhesion parameters of the inventive compositions fulfill the requirements of the EN 12004-1/2:2017 standard for the tile adhesives for internal use (adhesion tensile strength ≥ 0.5 N/mm^2). The measured values are higher or comparable that the ones of the non-inventive specimen of the same formulation, but without biochar.

The carbon footprint of the inventive composition, especially of the composition comprising higher amount of biochar, is reduced in comparison to the non-inventive adhesive composition.

Example 2

Table 3 shows a cementitious tile adhesive composition of the present invention. The composition has been prepared by intermixing all of the components in a dry state.

Table 3: Tile adhesives formulation of the present invention; all values given in [g].

Example	**2-1**
CEM I, 52.5 R	500
GGBS	529
Quarz sand, 0.1 – 0.3 mm	1211.94
Thickener	10
Alkaline activator	46
Biochar (0.1 – 0.4 mm)	100.59
Additives, etc.	192.61

The tile adhesive composition 2-1 has been mixed with water (weight ratio of water to total weight of dry adhesive composition was 0.4) in order to obtain a processable composition.

The obtained processable composition was then used for adhesion tests according to EN 12004-1/2:2017. The results of tensile adhesion strength extended open time and slip (measured according to standard EN 1308) are presented in Table 4.

The carbon footprint of the adhesive compositions disclosed in Table 3 was assessed based on the carbon sequestration potential of 2.58 g of CO_2 per g of biochar (corresponding to 85% fixed carbon in biochar). The Global Warming Potential (GWP) value of the inventive tile adhesive, including biogenic emissions, was 0.2744 kg $CO_{2\,eq.}$/ kg tile adhesive. The GWP was also calculated for the same formulation but comprising additional 100.59 g of sand instead of biochar and amounted to 0.3605 kg $CO_{2\,eq.}$/ kg tile adhesive.

Table 4: Result of inventive tile adhesive specimen (2-1).

Example	**2-1**
Initial tensile adhesion strength [N/mm^2]	1.27
Initial tensile adhesion strength after water immersion [N/mm^2]	1.21
Initial tensile adhesion strength after heat ageing [N/mm^2]	1.39
Extended open time: tensile adhesion strength	0.69
Slip (area weight: 50 kg/m^2) [mm]	8

The inventive cementitious tile adhesive composition fulfills the requirements of EN 12004-1/2:2017 for a C2TE tile adhesive. Additionally, its Global warming Potential, including biogenic emissions, is reduced by 24% in comparison to the same formulation but with more sand instead of biochar.

Example 3

Tables 5 and 6 show cementitious tile adhesive compositions (examples 3-1, 3-4, 3-5, and 3-8 are comparative examples not according to the invention, other examples are according to the invention). The compositions were prepared by intermixing all of the components in a dry state. Processable compositions were obtained by mixing the respective compositions with water in a weight ratio of water to powder as indicated in tables 5 and 6 respectively.

Tests of tensile adhesion were done according to Annex ZA, Table ZA.1.1 of EN 12004-1/2:2017. The coverage increase was measured as a reduction of the weight of processable material needed to cover a 1 m^2 surface when applied with a 6 mm notched trowel in comparison to the reference without biochar. Results are given in tables 5 and 6 below.

Table 5: Tile adhesives formulations and results

Example	**3-1**	**3-2**	**3-3**	**3-4**
CEM II/A-LL 52.5 R [g]	27	27	27	27
Cellulosic Ether HEMC [g]	0.25	0.25	0.25	0.25
RDP* [g]	0.7	0.7	0.7	0.7
Calcium formate [g]	0.5	0.5	0.5	0.5
Silica Sand [g]	71.55	69.55	67.55	65.55
Biochar (0.1 – 0.4 mm) [g]	0	2	4	6
Water/powder ratio	0.22	0.23	0.24	0.25
Coverage increase [%]	ref	5	10	14
Tensile adhesion after 28 d curing [MPa]	1.10	1.00	1.02	0.82
Tensile adhesion after water immersion [MPa]	0.77	0.70	0.62	0.55
Tensile adhesion after heat ageing [MPa]	0.97	0.91	0.81	0.73

* Based on vinylacetate-ethylene copolymer

Table 6: Tile adhesives formulation and results

Example	**3-5**	**3-6**	**3-7**	**3-8**
CEM II/A-LL 52.5 R [g]	22.0	22.0	22.0	22.0
GGBS** [g]	8.0	8.0	8.0	8.0
Cellulosic Ether HEMC [g]	0.3	0.3	0.3	0.3
RDP* [g]	1.2	1.2	1.2	1.2
Calcium formate [g]	0.8	0.8	0.8	0.8
NaCl [g]	1.0	1.0	1.0	1.0
Silica Sand [g]	66.7	64.7	62.7	60.7
Biochar (0.1 – 0.4 mm) [g]	0	2.0	4.0	6.0
Water/powder ratio	0.22	0.22	0.23	0.23
Coverage increase [%]	ref	8	13	17
Tensile adhesion after 28 d curing [MPa]	1.15	1.10	1.05	0.90
Tensile adhesion after water immersion [MPa]	1.10	1.10	1.05	0.95
Tensile adhesion after heat ageing [MPa]	1.15	1.05	1.00	0.85

* Based on vinylacetate-ethylene copolymer

** Blaine fineness 4000 g/cm^2

Example 4

Table 7 shows inventive cementitious tile adhesive compositions. The compositions were prepared by intermixing all of the components in a dry state. Processable compositions were obtained by mixing the respective compositions with water in a weight ratio of water to powder as indicated in table 7.

Tests of tensile adhesion were done according to Annex ZA, Table ZA.1.1 of EN 12004-1/2:2017. Slip was measured according to standard EN 1308. Wet density was determined by weighing a known volume of freshly prepared wet mortar.

Results are shown in below table 7.

Table 7: Tile adhesives formulation and results

Example	**4-1**	**4-2**	**4-3**	**4-4**
CEM I 52.5 R [g]	19.7	19.3	17.7	17.3
GGBS** [g]	20.8	20.4	21.4	20.9
RDP* [g]	2.5	2.5	2.5	2.5
Calcium formate [g]	1.9	1.9	1.9	1.9
Quartz Sand [g]	47.5	46.5	48.7	47.6
Biochar (0.1 – 0.4 mm) [g]	4	6	4	6
Expanded clay [g]	2	2	2	2
Admixture*** [g]	1.6	1.4	1.8	1.8
Water/powder ratio	0.39	0.40	0.39	0.40
Wet density [g/ml]	1.43	1.36	1.38	1.36
Slip [mm]	1	2	1	1
Tensile adhesion after 28 d curing [MPa]	1.11	1.20	1.12	1.07
Tensile adhesion, 30 min open time, after 28 d curing [MPa]	0.41	0.43	0.64	0.62
Tensile adhesion after heat ageing [MPa]	1.17	1.08	1.03	0.96

* Based on vinylacetate-ethylene copolymer

** Blaine fineness 4000 g/cm^2

*** Mixture of thickener (cellulose ether), water retention agent, cellulose fiber

Claims

1. A cementitious tile adhesive composition comprising:
   a) at least one cementitious binder;
   b) at least one aggregate;
   c) biochar,

   wherein, if the cementitious binder does not comprise slag, the weight ratio of biochar to the cementitious binder is between 0.001 and 0.15 to 1, preferably between 0.001 and 0.024 to 1, more preferably between 0.001 and 0.020 to 1, most preferably between 0.001 and 0.014 to 1, or

   wherein, if the cementitious binder comprises slag, the weight ratio of biochar to slag is between 0.16 and 0.5 to 1, preferably between 0.16 and 0.3 to 1, more preferably between 0.16 and 0.25 to 1, especially between 0.16 and 0.2 to 1.

2. The cementitious tile adhesive composition according to claim 1, characterized in that the cementitious binder is a composite cement of the type CEM II, CEM III, CEM IV or CEM V according to EN 197-1:2011, or is a limestone calcined clay cement.

3. The cementitious tile adhesive composition according to claim 1, characterized in that the cementitious binder comprises Portland cement and slag, preferably steelmaking slag or ground granulated blast furnace slag.

4. The cementitious tile adhesive composition according to claim 3, characterized in that the weight ratio of Portland cement to slag is between 0.2 and 2 to 1, preferably between 0.4 and 1.2 to 1, more preferably between 0.4 and 1 to 1, still more preferably between 0.8 and 0.99 to 1, especially between 0.8 and 0.9 to 1.

5. The cementitious tile adhesive composition according to at least one of the claims 1 – 4, characterized in that the biochar has an average particle size D50 of more than 0.1 mm, as determined by sieve analysis according to EN

12192-1:2002 or EN 933-1:2012.

6. The cementitious tile adhesive composition according to at least one of the claims 1 – 4, characterized in that the biochar has an average particle size D50 between 0.0001 mm and 0.5 mm, preferably between 0.1001 mm and 0.5 mm, more preferably between 0.1001 mm and 0.4 mm, most preferably between 0.2 mm and 0.4 mm, as determined by sieve analysis according to EN 12192-1:2002 or EN 933-1:2012.

7. The cementitious tile adhesive composition according to at least one of the previous claims, characterized in that the biochar is obtained by pyrolysis, preferably by slow pyrolysis, of woody biomass, herbaceous biomass, agricultural biomass, waste biomass, or manure biomass.

8. The cementitious tile adhesive composition according to any of the previous claims, characterized in that the biochar is obtained by pyrolysis at the temperature of at least 400 °C, preferably at least 500 °C, more preferably at least 600 °C, most preferably at least 700 °C.

9. The cementitious tile adhesive composition according to at least one of the previous claims, characterized in that it further comprises at least one alkaline activator, preferably alkali metal carbonate, alkali metal hydroxide, alkaline-earth metal hydroxide, alkaline-earth metal oxide, alkaline-earth metal formate, alkali metal silicate, alkali metal sulphate, alkali metal chloride or mixture thereof.

10. The cementitious tile adhesive composition according to at least one of the previous claims, characterized in that the surface of biochar is treated, preferably functionalized.

11. A processable composition comprising the cementitious tile adhesive composition according to any of the previous claims and water, wherein the weight ratio of water to cementitious tile adhesive composition is in the range from 0.2 to 0.5, preferably from 0.3 to 0.45.

12. Use of a cementitious tile adhesive composition according to at least one of claims 1 - 10 or of a processable composition according to claim 11 for tiles with surface dimensions of at least 300 x 300 mm, preferably at least 600 x 600 mm and/or for tiles with a water absorption of ≤ 0.5 % measured according to EN ISO 10545-3:2018, for example porcelain tiles.

13. A structure, in particular a floor, a wall or a ceiling, comprising a cover element, in particular a tile, whereby the cover element is fixed to the structure with a composition according to claim 11 which has hardened.

14. A method for preparing a cementitious tile adhesive composition with a Global Warming Potential of biogenic CO_2 emission of less than 0.07 kg $CO_{2\ eq.}$/ kg, preferably less than 0.065 kg $CO_{2\ eq.}$/ kg, more preferably less than 0.05 kg $CO_{2\ eq.}$/ kg, still more preferably less than 0.04 kg $CO_{2\ eq.}$/ kg, especially having a negative Global Warming Potential of biogenic CO_2 emissions, said method comprising a step of adding biochar to the cementitious tile adhesive composition.

INTERNATIONAL SEARCH REPORT

International application No
PCT/EP2025/071547

A. CLASSIFICATION OF SUBJECT MATTER
INV. C04B18/10 C04B28/04 C04B28/08 C04B18/14
ADD.

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)
C04B

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

EPO-Internal

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	EP 4 357 312 A1 (SAINT GOBAIN WEBER FRANCE [FR]) 24 April 2024 (2024-04-24) cited in the application claims 1,4,6,9-10,12 example 1 paragraphs [0018], [0022], [0035], [0047], [0050], [0054] -----	12,13
X	US 2020/062646 A1 (NG MAY SHUAN [SG] ET AL) 27 February 2020 (2020-02-27) claims 1,3-4,6-10 paragraphs [0001], [0064], [0066], [0146] table 2 example 1 figure 1 -----	1,2,5,7, 8,14

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[x] Further documents are listed in the continuation of Box C.

[x] See patent family annex.

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Date of the actual completion of the international search	Date of mailing of the international search report
4 November 2025	12/11/2025

Name and mailing address of the ISA/	Authorized officer
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INTERNATIONAL SEARCH REPORT

International application No
PCT/EP2025/071547

C(Continuation). DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
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A	US 2023/047807 A1 (MAGUIRE MARC [US] ET AL) 16 February 2023 (2023-02-16) claims 1,5,20 examples 2-4,6 -----	1-14
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